



Keerthana Vino Kumar | SDET/QA Analyst

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SUMMARY

ISTQB-certified QA Analyst with over 11 years of experience in manual and automation testing across web, desktop, and mobile platforms. Proven expertise in designing and implementing scalable test automation frameworks using Selenium WebDriver, TestNG, resulting in significant reductions in manual effort and production defects. Skilled in functional, regression, integration, exploratory, backend, accessibility, and performance testing, with a strong focus on quality, compliance, and business alignment. Hands-on experience in CI/CD pipeline integration with tools such as Jenkins, GitHub Actions. Strong background in accessibility testing aligned with WCAG and Section 508 using NVDA and ANDI, achieving measurable reductions in ADA violations. Led QA efforts in Agile teams by translating user stories into detailed test cases, strategies, and RTMs. Created automated and manual test cases for critical modules. Regularly collaborated with cross-functional teams, participated in sprint planning, defect triage, and UAT, and mentored junior engineers on automation frameworks and best practices. Successfully reduced post-release defects and hotfixes by implementing effective test strategies and root cause analysis using SQL and log tracing. Recognized for improving release stability, enhancing automation reliability, and maintaining audit readiness through structured test documentation and traceability.

SKILLS

Test Automation: Selenium WebDriver, Playwright

Test Frameworks: TestNG, JUnit, Cucumber (BDD), Maven Surefire, BDD with Gherkin

Languages: Java, SQL

API Testing: Postman, REST Assured

CI/CD Tools: Jenkins, GitHub Actions

Version Control: GitHub, Git

Accessibility Testing: WCAG, Section 508, NVDA, ANDI, TalkBack

Test Management: JIRA, Zephyr

Databases & ETL: SQL, SSIS

Testing Types: Functional, Regression, Integration, API, Performance, Security, Accessibility, Mobile, Backend, Data-Driven

Others: RTM Creation, UAT Support, ETL Validation, Data Migration Testing

CERTIFICATIONS

- ISTQB Foundation Level - International Software Testing Qualification Board
- Mobile App Tester - American Software Testing Qualification Board (ASTQB)

PROFESSIONAL EXPERIENCE

Client: Department of Labor, Remote | Nov 2022 – Till date

Senior SDET/QA Analyst

- Analyzed Business Requirements Documents (BRDs) and Functional Specifications (FSDs) to create detailed test strategies and ensure full requirements coverage.
- Translated business and functional requirements into technical documentation and high-level test design artifacts.
- Participating in requirement sessions, reviewing key features, and documenting processes to ensure thorough QA coverage for the Department of Labor's ROAP Project.
- Efficiently managed defect resolution, excelling in error identification, feedback provision, and root cause analysis.
- Liaise with federal and state product owners to optimize product quality through improved understanding of their operational frameworks.
- Accountable for test script creation, execution and UAT demonstrations.
- Responsible for Section 508 accessibility testing, SMOKE testing and happy path regression.

- Investigate Production Issues referred by technical support.
- Providing support for each region rollout for user guide, training videos and end user training demonstration.
- Performed functional, regression, integration, exploratory, backend, performance, usability, and GUI testing across web and mobile applications.
- Designed BDD test frameworks using Cucumber, integrated with Selenium WebDriver, Jenkins, and GitHub for CI/CD automation.
- Conducted API and backend testing using REST Assured, Postman ensuring data integrity and system interoperability.
- Extensive experience in SQL query writing and executing.
- Designed and implemented Playwright automation framework (Java + TestNG) for regression and smoke testing, ensuring faster and stable test executions.
- Integrated Cucumber BDD with Playwright to write human-readable test cases, improving collaboration with business stakeholders.
- Automated end-to-end UI flows including login, form submissions, dashboard validations, and document uploads using Playwright.
- Handled parallel execution using Playwright browser contexts, reducing regression execution time by 40%.
- Created and executed functional and UI test cases, increasing test coverage and reducing post-release bugs.
- Conducted accessibility testing using WCAG, Section 508, NVDA, and ANDI, reducing ADA violations by 25%.
- Identified and resolved over 300+ defects in JIRA, improving defect resolution cycle time by 18%.
- Collaborated closely with product owners, developers, and UAT teams to ensure seamless releases and reduce hotfixes by 30%.

Client: Walgreens, Chicago, IL | Apr 2017 - Oct 2022

SDET/QA Engineer (Automation & Manual)

- Gathered and analyzed business and technical requirements to define QA scope and ensure test coverage for high-traffic ecommerce platforms.
- Translated user stories into well-defined test cases, Test Plans, Test Strategies, RTMs, and comprehensive test data sets.
- Designed and executed manual and automated test cases across key modules: product catalog, shopping cart, checkout, order management, and payments.
- Performed smoke, functional, integration, regression, backend, and data-driven testing across web and mobile platforms.
- Authored BDD scenarios using Gherkin syntax for use with the Cucumber framework, improving test clarity and stakeholder communication.
- Coordinated closely with DevOps and Product teams to align test data with real-world scenarios and user behaviors.
- Conducted data-driven testing using Excel and SQL, validating business rules such as dynamic pricing and inventory updates.
- Tracked user stories, test cases, and defects using JIRA, maintaining visibility and accountability throughout the sprint cycle.
- Built a hybrid automation framework using Selenium, Java, and TestNG, reducing manual regression cycles by 40%.
- Developed and executed REST API validations across backend services to ensure data consistency and real-time updates.
- Maintained version-controlled automation scripts in GitHub, improving collaboration and change traceability.
- Participated in sprint planning, daily stand-ups, and defect triage meetings to ensure QA alignment with Agile delivery cycles.
- Integrated automation into CI/CD pipelines using Jenkins and TeamCity, enabling real-time test feedback during deployments.

Client: First Data (Acquired by Fiserv), Chicago, IL | Sep 2016 – Mar 2017

QA Analyst

- Validated end-to-end financial transaction data from POS systems, ensuring backend data integrity across Oracle and AWS DocumentDB databases.
- Authored and executed 100+ manual test cases based on business requirements and user stories for core banking workflows.
- Performed complex SQL validations to verify financial computations, interest calculations, and transaction records.

- Conducted comprehensive functional, regression, integration, API, mobile, and backend testing. ● Integrated automated test suites with Jenkins and Azure DevOps, streamlining CI/CD workflows in Agile release cycles.
- Collaborated with DevOps teams to validate smoke/sanity tests post-deployment, reducing rollback incidents by over 20%.
- Automated test result tracking and reporting in JIRA, improving traceability and accountability during financial sprints.
- Led test planning and execution using Zephyr, aligning QA milestones with project deadlines and regulatory goals.
- Logged and managed defects in HP Quality Center, establishing full Requirements Traceability Matrix (RTM) for audit readiness.

Client: AT & T and GSK, Pfizer/BAT, India | Aug 2011 - Mar 2015

Implementation: Infosys

Test Engineer (Manual + Automation)

- Performed manual testing for web and desktop applications across multiple modules. ●
- Conducted accessibility testing using NVDA and TalkBack as per WCAG 2.0 and Section 508. ●
- Created and executed test cases based on functional requirements.
- Participated in defect triage and retesting activities with developers.
- Supported UAT cycles to validate key user flows and accessibility compliance.
- Validated data flows using SQL scripts and SSIS; reduced ETL-related defects by 20%. ●
- Provided QA support for 5+ releases, maintaining 100% uptime post-deployment.
- Created SQL validation packs for backend data testing across multiple environments. ●
- Participated in QA environment setup and supported smooth transitions during system migrations. ●
- Supported backend QA processes for large-scale data migrations, ensuring clean hand-offs with zero post-release issues.

EDUCATION

- Bachelor of Engineering (Computer Science) Anna University, India Sep 2007- Apr 2011